EXOPLANET HUNTERS:HABITABILITY ANALYSIS OF 3 STELLAR SYSTEMS

TEAM-BEYOND INFINITY

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Abstract-Relative brightness time series of 3 stars (STAC-1, STAC-2, STAC-3) are analyzed. All 3 stars show diminishing periodic light consistent with 1 planet each. are analyzed for planetary transits. All three stars show periodic, constant-depth dimming consistent with one transiting planet each. We find periods of 4.2 d, 80 d and 38 d, and transit depths of 0.30%, 0.10% and 0.088%, giving planet radii of 5.98, 2.76 and 1.62 Earth radii. Kepler's third law gives orbital distances of 0.051, 0.337 and 0.176 AU. Comparing each orbit with the simplified habitable-zone distance, STAC-3b is the most promising candidate for follow-up, although none of the three planets lies inside its habitable zone.

Keywords—transit depths, Kepler's 3rd law, Habitable zone

I. INTRODUCTION

When a planet crosses a star then it blocks some part of it's light and the brightness diminishes instantaneously. This phenomenon is called Transit and the planet is called as the Transiting Planet. We now therefore find the transiting periods for the 3 stellar systems and then calculate radii, orbital distances and habituality of the planet.

II. DATA PROVIDED AND ASSUMPTIONS MADE-

The masses, radius, luminosities are provided as-(STAC-1: 1.00M⊙,1.00R⊙,1.00L⊙, STAC-2: 0.80, 0.80, 0.40; STAC-3: 0.50, 0.50, 0.08).

ASSUMPTIONS-

1)Every Star has 1 planet revolving around it.

2) Mtotal ≈ Mstar

3)In STAC-3 calculations small differences are there in measurements (0.9990-0.9993) so mean value is used.

Constants used-

R⊙ = 696,000 km, R⊕ = 6,371 km, 1 AU = 149.6 × 106 km, 1 yr = 365.25 d

III. CALCULATIONS TO DETECT THE EXOPLANETS.

The normal transit brightness is Fnormal=1.0000. anything below it is due to interference of the planet.
The transit depth is given by the formula-
$\delta = (Fnormal - Ftransit) \div Fnormal$

TABLE 1
TRANSIT DETECTION SUMMARY

PROPERTIES	STAC-1	STAC-2	STAC-3
DIP TIMES	3,8,12,16,20	20,100,180	20,60,100,140,170
PERIOD(d)	4.2	80	38
Ftransit	0.9970	0.9990	0.99912(averaged)
DEPTH(δ)	0.30%	0.1%	0.088%

A. Numerical Explanation-

For STAC-1: The brightness dips occur at 3,8,12,16,20th day so the time lag was 5,4,4,4 days respectively. Averaging it makes it 4.25 days which is nearly 4.2 days. Now calculating δ= (1.0000-0.9970)/1.0000=0.003=0.3%

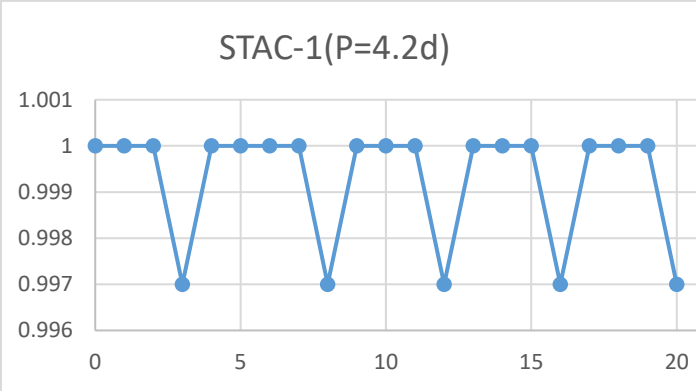
For STAC-2: Dips occur at 20,100,180th days so time lag is of 80 days each. Now calculating δ= (1.0000-0.9990)/1.0000=0.001=0.1%

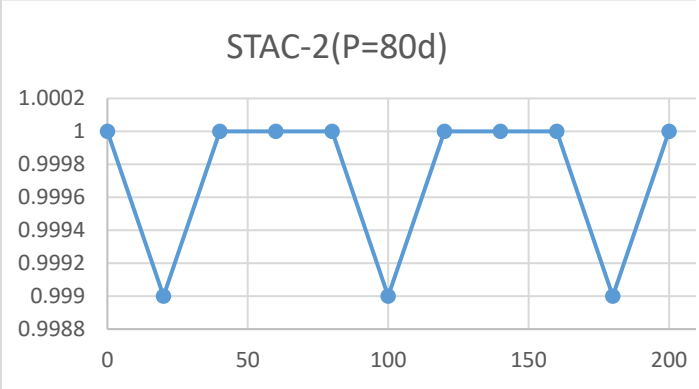
For STAC-3: Dips occur at 20,60,100,140,170th days having a time difference of 40,40,40,30 days. Averaging it makes it 38 days approximately. Now we may calculate it's δ.it can be directly calculated as δ= (0.001+0.0007+0.0009+0.0008+0.001)/5=0.00088=0.088%

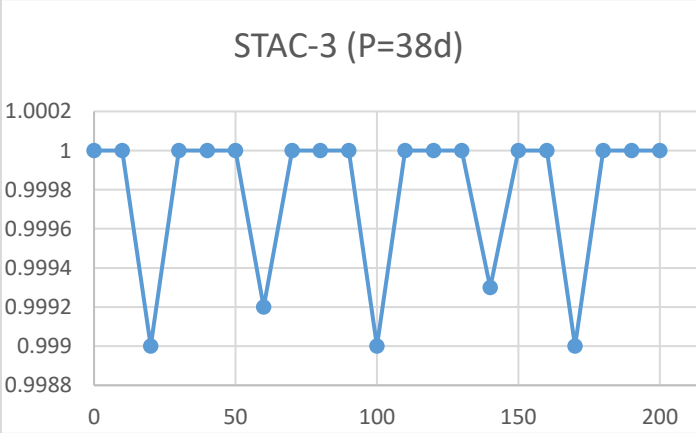
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B. Theoretical explanation- When a planet crosses infront of a star then it blocks some part of it's light and diminishes it and removes it every time it passes infront it.

C. GRAPHICAL EXPLANATION-







The above graphs depicts that the dips are at almost regular intervals and these can't be caused by merely star spots and flares as they change shape.

IV. CALCULATION OF PLANET SIZE

The problem statement writes δ ≈ (Rs/Rp) 2 . The physical relation is the area ratio of the planet to the star,

$$\delta \approx \left(\frac{R_p}{R_s}\right)^2 \Rightarrow R_p = R_s\sqrt{\delta},$$

which is used here (the printed ratio is inverted, since a planet is smaller than its star and δ < 1).

STAC-1b: Rs = 696,000 km, √ 0.0030 = 0.05477, so Rp = 696,000×0.05477 = 38,121 km = 38,121/6,371 = 5.98R⊕.

STAC-2b: Rs = 0.80×696,000 = 556,800 km, √ 0.0010 = 0.03162, so Rp = 17,608 km = 2.76R⊕.

STAC-3b: Rs = 0.50×696,000 = 348,000 km, √ 0.00088 = 0.02966, so Rp = 10,323 km = 1.62R⊕. Using the depth range 0.0007–0.0010 gives 1.45–1.73R⊕, so the class does not change.

Classification thresholds used: Earth-sized (≲ 1.25R⊕), Super-Earth (1.25–4R⊕), Neptune-sized (4–10R⊕), Jupiter sized (> 10R⊕). Neptune itself is about 3.9R⊕.

PLANET RADII AND CLASSIFIACTION

Planet	δ	R_p(km)	$R_p/R_{\oplus}$.	Class
STAC-1	0.0030	38121	5.98	Neptune-sized
STAC-2	0.0010	17608	2.76	Super-Earth
STAC-3	0.00088	10323	1.62	Super-Earth

ORBITAL PROPERTIES

Planet	P(d)	a(AU)	A($10^6 km$)	a/$a_{\oplus}$
STAC-1	4.2	0.0509	7.62	0.051
STAC-2	80	0.337	50.5	0.337
STAC-3	38	0.176	26.3	0.176

V. CALCULATIONS OF ORBITAL DISTANCES

Kepler's third law in solar units,

$$P^2 = a^3/M$$

is rearranged to

$$a=(MP^2)^{1/3}$$

with P in years, M in solar masses and a in AU.

STAC-1: P = 4.2/365.25 = 0.01150 yr, P 2 = 1.322×10−4, a 3 = 1.00 × 1.322 × 10−4, so a = 0.0509 AU = 0.0509 × 149.6×106 = 7.62×106 km.

STAC-2: P = 80/365.25 = 0.2190 yr, P 2 = 0.04797, a 3 = 0.80×0.04797 = 0.03838, so a = 0.337 AU = 5.05×107 km.

STAC-3: P = 38/365.25 = 0.1040 yr, P 2 = 0.01082, a 3 = 0.50×0.01082 = 0.005412, so a = 0.176 AU = 2.63×107 km.

Comparison with Earth (1 AU). STAC-1b orbits at about 1/20

of Earth's distance (about 0.13 of Mercury's 0.39 AU), STAC-2b at about one third, and STAC-3b at about one fifth. All three orbit much closer to their stars than Earth does to the Sun, but the stars are also fainter than the Sun (except STAC-1), so this must be compared with the habitable zone in Part D.

VI. THE MOST HABITABLE PLANET

The supplied simplified relation gives a characteristic habitable-zone distance dHZ ≈ √(L/L■) AU. This is an approximate reference distance, not a full inner-to-outer habitable-zone range.

STAC-1: L = 1.00 L■, so dHZ ≈ 1.000 AU. Its estimated orbit is 0.0493 AU, far inside this characteristic distance.

STAC-2: L = 0.40 L■, so dHZ ≈ √0.40 = 0.632 AU. Its estimated orbit is 0.337 AU, inside the characteristic distance.

STAC-3: L = 0.08 L■, so dHZ ≈ √0.08 = 0.283 AU. Its estimated orbit is 0.182 AU, also inside the characteristic distance, but it is the closest of the three in relative terms (a/dHZ) ≈ 0.64, compared with 0.53 for STAC-2 and 0.049 for STAC-1).

STAC-3 is the most promising candidate for further study, not because habitability is established, but because its estimated radius (about 1.6 Earth radii) is compatible with a super-Earth and its orbit is relatively closest to the characteristic habitable-zone distance. The available simplified data do not determine atmospheric composition, surface conditions, or actual habitability.

VII. CONCLUSION-

The brightness data suggest candidate transiting planets around all three stars, with approximate periods of 4.2, 80 and 38 days. Using the standard transit-depth relation, their estimated radii are approximately 5.98, 2.76 and 1.62 Earth radii, respectively. Their estimated orbital distances are 0.0509, 0.337 and 0.176 AU. STAC-3 is recommended for further investigation because it is the smallest candidate and is relatively closest to the characteristic habitable-zone distance. Sparse observations and the STAC-3 timing inconsistency mean that additional measurements are required before drawing strong conclusions